

Modular Programming 2-oyligi

2 - oy uchun plan

- 1) 4 ta Project (data extraction + Deployment)
- 2) Modelni rivojlantirish => (tekshirish, neyronlar)
- 3) Modular Programmingni ham rivojlantiramiz (structure)
- 4) Git bash commands
- 5) Multi-class classification
- 6) Multi-output models
- 7) Deployment

Asosiy qobiliyatlarni shakllantirish + rivojlantirish:

- Problem solving (codewars or leetcode) ^{at least one problem each day}
- To'g'ri va tez yozish (monkeytype)
- Logical thinking (kitob o'qish) (olinda bo'li)

Multi-class classification

- Binary Classification => bu 2 ta classificationdan iborat bo'ladi

↳ (Yes or No, 0 or 1, false or True)

Binary Classification

No.

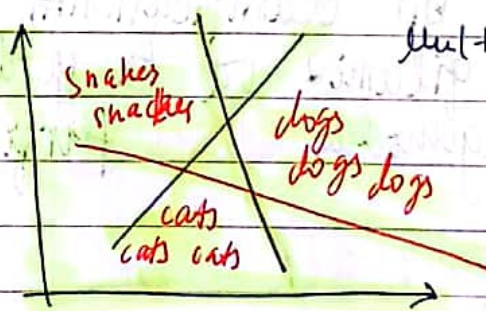
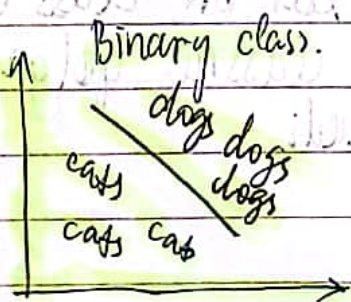
- 2 ta class li target ofiyomat mavjud bo'lgan dataset k.k bo'ladi.
- Confusion matrixni tushunish oson (0/1)
- 0/1 ligi uchun modelni tushunish oson.
- Faqat bitta classifier modelni talab qiladi.
- One vs another (0/1, yes/no)

OVA

Multi-class classification

- Logistic Regression (oqasida sigmoid function or logistic function (0 va 1 oralig'ida)) → Binary Classification (DT, RF → Binary and Multi class)

- Xaqiqiy xayotda ko'plab xolatlar multiclass



- data kutta va balansli bo'lish k.k

Yechim

- Multi-classlarni ko'plab binaryga xolatga bo'lib olish.

• 2 ta asosiy strategiya mavjud:

— One-vs-Rest (OvR)

0 1
it vs (ilon + mushuk)

0 1
mushuk vs (it + ilon)

0 1
ilon vs (it + mushuk)

— One-vs-One (OvO) $\Rightarrow$ usulida har bir klass boshqa har bir klass bilan alohida model orqali solishtiriladi.

1. cat vs dog

2. cat vs snake

3. snake vs dog

$$\geq \frac{k(k-1)}{2}$$

• OvR

— Biz bir classification modelni xar bir class u-n train qilamiz va bu yerda u bir classni qolgan/ini yig'ilganlardan (rest) farqini o'rganadi.

misol: 3 ta class

— ilon

— It

— mushuk

• Model 1: ilon vs (It + mushuk)

• Model 2: It vs (ilon + mushuk)

• Model 3: mushuk vs (It + ilon)

Agar classimiz N ta bo'lsa N marta train qilamiz.

By yerda biz 3ta modelarni train qilamiz (1ta algorithm)

OvR

↳ Prediction

↳ Model 1 : $\text{dlon} \text{ vs } (\text{It} + \text{llushuk}) \rightarrow 0.65$

↳ Model 2 : $\text{It} \text{ vs } (\text{dlon} + \text{llushuk}) \rightarrow 0.18$

↳ Model 3 : $\text{llushuk} \text{ vs } (\text{It} + \text{dlon}) \rightarrow 0.25$

↓
Predicted class : dlon

OvR : Example 1

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.multiclass import OneVsRestClassifier,
    OneVsOneClassifier
from sklearn.metrics import classification_report
```

```
X, y = load_iris(return_X_y=True)
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42)
```


`ovr_model = OneVsRestClassifier(LogisticRegression(
max_iter = 1000, random_state = 42))`

`ovr_model.fit(X_train, y_train)`

Predict and evaluate

```
y_pred_ovr = ovr_model.predict(X_test)
print("=== One-vs-Rest (OvR) ===")
print(classification_report(y_test, y_pred_ovr))
```

OvR + (advantages)

- ↳ Sodda va oson ishlatish
- ↳ Agar classlar ozroq bo'lsa tez ishlaydi.

OvR - (disadvantages)

- ↳ Imbalanced dataset/ b/n yaxshi ishlamaydi.

OvO : One-vs-One Classifier

Bu yerda biz classification modellarni har bir class uchun **train qilamiz** va bu yerda har bir class'ni **qolganlari bilan yakkama yakka farqini o'rganamiz**.

misol : 3 ta class

- Iton
- It
- Ilushuk

Model 1: Iton vs it

Model 2: Iton vs ilushuk

Model 3: It vs ilushuk

OvO: Example 2

```
ovo_model = OneVsOneClassifier(LogisticRegression(
    max_iter = 1000, random_state = 42))
```

```
ovo_model.fit(X_train, y_train)
```

```
y_pred_ovo = ovo_model.predict(X_test)
```

```
print("=== One-vs-One (OvO) ===")
```

```
print(classification_report(y_test, y_pred_ovo))
```

OvO formulasi: $\frac{n(n-1)}{2}$

OvO vs OvR farqning asosiy farqi

Modellar sonida va sekin ishlashida

→ OvO :

↳ Prediction

↳ Model 1: Olon vs it (ilan)

↳ Model 2: Olon vs mushuk (mushuk)

↳ Model 3: It vs mushuk (mushuk)



• Predicted class: Mushuk

• Agar teng bo'lib golsa confidence scorega ko'ra eng yuqori score dyanı predict qilinadi.

• Confidence $\Rightarrow$ Estimollar nazariyasi

confidence score formulasi

→
$$\text{Con. Score} = \frac{1}{1 + e^{-z}} = (z \approx 2.71)$$

$$= \frac{e^z}{e^z + 1} \Rightarrow (0, 1) \text{ oralig'ida bo'ladi.}$$

OvO + (advantages)

↳ Imbalanced dataset da ishlatish yaxshi

↳ Hecicem classlarda xam yaxshi

OvO - (disadvantages)

• 10 ta classes $\rightarrow$ 45 models $\frac{n(n-1)}{2} = \frac{10(10-1)}{2} = 45$

• 100 ta classes $\rightarrow$ 4950 ta models

• Nimaga olib keladi? $\Rightarrow$ 1. Ortig'cha chiqim, sarf-gor olib keladi. 2. Dataset ko'p bo'lsa ko'p vaqt ketadi. 3. Modelni sekin ishlatishga ham olib keladi.

Bu shunchaki biz train qilayotganimizda, lekin hali model tayyorlab olgandan keyin, uni xavfli ishga tushirayotgan paytimizda ham buni monitoring qilish, buni natijasini olishimiz ozinga qiyinchilik tuqdiradi.

Lekin baribir aytib o'tilinganidek bizni MLda algoritim ko'ramizmi biron bir usul, method ko'ramizmi, qaysi birida qiymat yozishi chiqishi u-n albatta ko'proq vaqt talab qiladi.

~~chunki qozon~~

Multi-class classification umumlashgan holda

- Ko'plab class outputga ega bo'lgan dataset k-k bdi.
- Multi-class model:

- One-vs-Rest (OvR)

• Har bir model bitta classni boshqalardan ajratib olishga harakat qiladi (yes or no).

• Example: Nihata class bo'lsa shuncha modellarni train qilamiz.

- One-vs-One (OvO)

• Har bir ~~class~~ model bitta classni keyingi class bilan taqqoslab o'rganadi.

• Example: $n(n-1)/2 \Rightarrow 4(4-1)/2 = 6$ ta model train qilindi.

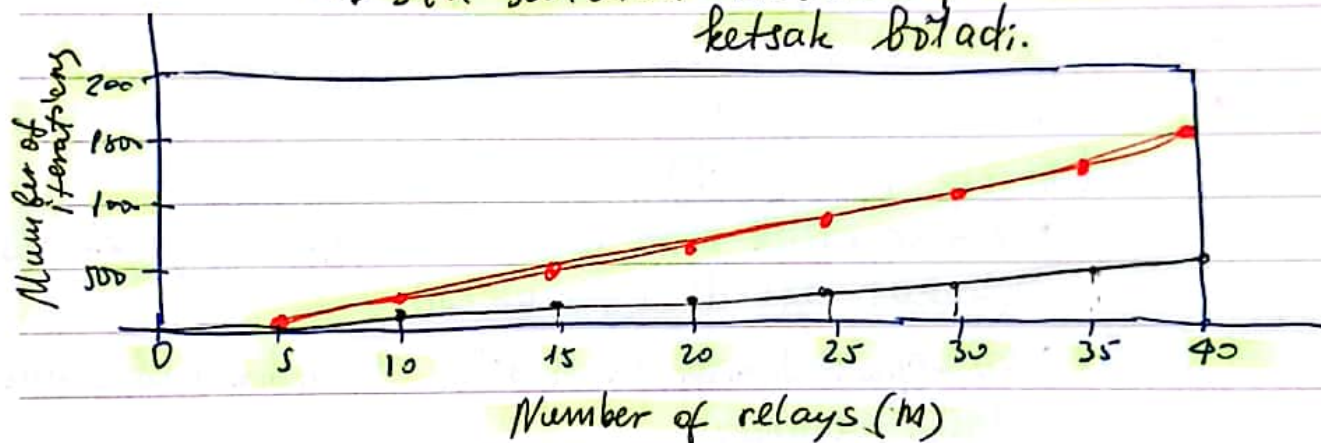
Max - iterations (parametr)

- Manziliimizga ~~h-h~~ ~~h-h~~ gadam ba - gadam borish uchun gadam $\rightarrow$ Best solution
- Bazida oz bazida kōp gadam h-h bo'ladi.

Algorithms :

- Linear (ML da)
- DL (Deep Learningda)
- Clustering (Unsupervised ML da)

↳ 3ta xolatda max-iter parametrni ishlatil ketsak bo'ladi.



Eng yaxshi tushuncha:

- Iteration = model = weightlarini yaxshiroq tomonga bir marta yangilashi.
 - Optimization = eng yaxshi weightlarni topish jarayoni
- max-iter = shu jarayonda modelga berilgan maksimal urinishlar soni.

Logistic Regression (max-iter = 1000)

"Model, eng yaxshi weight/ni topish u-n 1000 marta o'ch urinishing mumkin"

Jika master'ni bermasak otomatis
(default = 100) oleh ketadi.

• Kiri: Kiri
• Kiri: Kiri
• Kiri: Kiri

